

Problem A. Everfree Forest

Input file: standard input
Output file: standard output
Time limit: 1 second
Memory limit: 256 megabytes

The Everfree Forest, located in the northwestern Equestrian Wasteland, is a large expanse of woodland that has yet to be exposed to megaspell radiation.

To learn Everfree Forest from a scientific perspective, Twilight Sparkle considers it as an **undirected connected** graph of n vertices **without self-loop and multiple edges**, the degree of vertex in which is no more than 3.

After field exploration, Twilight Sparkle believes there are exactly k 3-degree vertices in this graph.

Now she wants to know, what's the maximal edges the graph could have, and what about the minimal edges?

Input

The first line contains a single integer T ($1 \leq T \leq 10^5$) - the number of testcases.

Each testcase contains only one line, two integers n, k ($0 \leq k < n, 6 \leq n \leq 10^5, \sum n \leq 10^5$).

Output

For each testcase:

The first line contains a single integer m_1 - the maximal edges it could have.

The next m_1 lines, each line contains two integers u_i, v_i - the edges in it.

The next line contains a single integer m_2 - the minimal edges it could have.

Then next m_2 lines, each line contains two integers w_i, p_i - the edges in it.

You should make sure that the graph you output is an **undirected connected** graph of n vertices **without self-loop and multiple edges**.

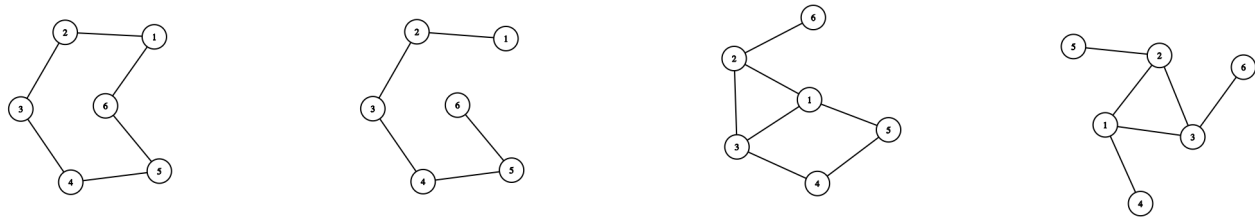
If there are multiple solutions, just output any of them, it could be proved that there exists at least one solution.

Example

standard input	standard output
2	6
6 0	1 2
6 3	2 3
	3 4
	4 5
	5 6
	6 1
	5
	1 2
	2 3
	3 4
	4 5
	5 6
	7
	1 2
	2 3
	3 4
	4 5
	5 1
	1 3
	2 6
	6
	1 2
	2 3
	3 1
	1 4
	2 5
	3 6

Note

There are valid solutions of the test case.



maximal edges in testcase 1 minimal edges in testcase 1 maximal edges in testcase 2 minimal edges in testcase 2

Problem B. a math problem

Input file: **standard input**
Output file: **standard output**
Time limit: 2 seconds
Memory limit: 512 megabytes

This is an easy math problem.

We define a function $f(x)$ to represent how many pairs of factors (p, q) of x satisfy that p and q are coprime. When $p \neq q$, (p, q) and (q, p) are different pairs.

$$f(x) = \sum_{p, q | x} [gcd(p, q) = 1]$$

Now we want to calculate

$$g(n) = \sum_{i=1}^n f(i)$$

The answer maybe too large, so you only need to output the result of its modulo of 998244353

Input

A single integer $n (1 \leq n \leq 10^{10})$

Output

A single integer $g(n)$

Example

standard input	standard output
10	48

Problem C. Catch

Input file: `standard input`
Output file: `standard output`
Time limit: 1 second
Memory limit: 256 megabytes

A tree is an undirected connected graph with n nodes and $n - 1$ edges. A leaf node refers to a node with degree 1, that is, a node connected to only one node.

You are given an unrooted tree, there are hamsters in some nodes of this tree, and your task is to catch all the hamsters. For this, you can place mousetraps at some nodes. If the hamster moves to the node where the mousetrap is placed, it will be caught immediately. Note that in this problem we consider the capacity of the mousetrap to be infinite, that is, the mousetrap at a single node can also hold an infinite number of hamsters.

Every second, each hamster moves to an adjacent vertex. A hamster cannot move along the same edge twice in a row, unless it entered a leaf node from a non-leaf node in the previous step, in which case it can return from the original edge. Find the minimum number of nodes to place mousetraps so that all hamsters will eventually be caught, regardless of their initial positions and escape strategies.

Input

The first line of input contains a positive integer $n(1 \leq n \leq 10^5)$, representing the number of nodes in the tree.

The following $n - 1$ lines, each line contains 2 positive integers $u, v(1 \leq u, v \leq n)$, indicating that there is an edge connecting node u and node v .

Output

Print a positive integer representing the minimum number of nodes on which to place the mousetrap.

Examples

standard input	standard output
3 1 2 2 3	1
10 2 8 5 10 3 4 1 6 3 9 1 7 4 8 10 8 1 8	2

Note

For the first testcase, you can place a mousetrap at node 2, because every possible route of the hamsters will certainly include node 2.

Problem D. Coffee

Input file: standard input
Output file: standard output
Time limit: 3 seconds
Memory limit: 512 megabytes

The girls of HTT like drinking tea. But one day, they wanted a change and decided to try coffee in the next n days.

Now Mugi, who always provides food and drinks for HTT, will go to the shop to buy some coffee. She learns that the price of coffee on day i is a_i . Now she has m coupons. The i th coupon can be used before day r_i (it can also be used on day r_i) and can reduce the price of coffee on that day by w_i . **Notice that the price can be a negative number after using the coupon.** Each coupon can be used only once, and Mugi cannot use more than one coupon per day.

Since the girls of HTT still need to drink tea, Mugi decided to choose **exactly** k days to buy coffee. Now she wants to know the minimum money she will spend (or the maximum money she can get) to choose exactly k days to buy coffee.

Input

The first line contains two integers n, m ($1 \leq n \leq 10^5, 1 \leq m \leq 2 \times 10^5$) - the number of days and the number of coupons.

The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^9$), and a_i denoting the price of coffee on day i .

Next m lines, each line contains two integers r_i, w_i ($1 \leq r_i \leq n, 1 \leq w_i \leq 10^9$), denoting that the i th coupon can be used before day r_i and can reduce the price by w_i .

Next line contains an integer k ($1 \leq k \leq n$), denoting the number of days that Mugi will buy coffee.

Output

One integer - the minimum money Mugi will spend to buy coffee. Notice that the answer can be a negative integer.

Examples

standard input	standard output
5 2 1 2 3 4 5 3 1 4 2 5	12
7 3 4 3 1 10 3 8 6 4 9 3 8 4 5 5	-5

Note

In the first example, Mugi can use coupon 1 on day 1 and use coupon 2 on day 2 to spend $0 + 0 + 3 + 4 + 5 = 12$.

In the second example, Mugi can use coupon 1 on day 2, coupon 2 on day 3 and coupon 3 on day 1. Then she can choose to buy coffee on day 1, 2, 3, 5, 7 and spend $(-1) + (-6) + (-7) + 3 + 6 = -5$.

Problem E. A Counting Problem

Input file: `standard input`
Output file: `standard output`
Time limit: 1 second
Memory limit: 256 megabytes

Find the number of x satisfied the following condition:

- $x \in [0, 10^n)$
- $2^a \times 3 \mid x$

Input

The first line contains a single integer t ($1 \leq t \leq 10000$) — the number of testcases. Each testcase contains two integers n and a in a line. ($1 \leq a \leq n \leq 10^{18}$)

Output

For each testcase, print the number of x satisfied the given conditions in a single line. Since the answer can be large, print it modulo 998244353.

Example

standard input	standard output
1	2
1 1	

Problem F. Fighting in Group

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

There are n cats here, numbered from 1 to n . They desire to fight.

We call the cat numbered i as “Cat i ”. For **every** $i \in [1, n]$, Cat i wants to fight with every **nonadjacent** cat.

For example, when $n = 4$, Cat 1 wants to fight with 3 and 4, Cat 2 wants to fight with 4, Cat 3 wants to fight with 1, Cat 4 wants to fight with 1 and 2.

You should arrange rounds to satisfy their desires. In each round, you can choose some cats from them, and divide the chosen cats into two nonempty groups. Obviously, a cat can't appears in both groups in a single round.

And in this round, **every** pair (i, j) which Cat i and j is chosen and set in different groups will fight with each other.

Be careful, you **cannot** arrange adjacent cats to fight in any round. They will feel unhappy. Formally, if you want to choose Cat i and Cat $i + 1$ in a round, they should in the same group.

For example, you can arrange $[1, 2, 4]$ to fight with $[6, 7]$. But you can't arrange $[1, 2, 4]$ to fight with $[3, 6, 7]$, because 3 don't want to fight with 2 or 4.

We don't care about how many fights actually happened in all rounds. In other words, for each pair (i, j) when $|i - j| > 1$, they should fight at least once in all rounds. Arrange two cats to fight again and again is not banned, although it may be useless.

To save time, you should find the **minimum** number of rounds to satisfy them all, and how to arrange each round.

If there are mutiple solutions, output any of them.

Input

The input contains an integer n ($3 \leq n \leq 10^3$) — the number of cats waiting for fight.

Output

The first line contains an integer m — the number of rounds you arrange.

The following $2m$ lines shows each round you arrange. For each round, print two lines for your two groups:

The first line contains n_1 — the number of cats in first group, and followed with n_a integers $a_1, a_2, \dots, a_{n_1}$, donates the cat's id.

The second line is similar, contains n_2 and $b_1, b_2, \dots, b_{n_2}$.

Examples

standard input	standard output
3	1 1 1 1 3
4	2 2 1 2 1 4 1 1 1 3

Note

In the first example, we only need one round to let Cat 1 and Cat 3 fight with each other.

In the second example, we can prove 2 is the minimum rounds we need. $(1, 3)$, $(1, 4)$, $(2, 4)$ needs to fight in this example. And there are many correct solutions for $n = 4$. The following solution is considered correct too:

2

1 1

2 4 3

2 1 2

1 4

Problem G. MicrosoftHearts

Input file: standard input
Output file: standard output
Time limit: 2 seconds
Memory limit: 512 megabytes

Microsoft Hearts is a classical game of Microsoft Windows. Let's look at *Microsoft Hearts* of 2 players.

Each player has n cards initially. There is one of the four suits on one single card: H-Heart♥, S-Spade♠, D-Diamond♦, C-Club♣.

The cards are ranked by the point on it. The smallest is 2, and the greatest is A, in which A stands for 14, T stands for 10, i.e. $2 < 3 < 4 < 5 < 6 < 7 < 8 < 9 < T < J < Q < K < A$.

The game has following rules.

Each round, the player who has the token would put one of his/her card out. Then the other player can play as following restrictions.

1. If the player has a card of the same suit of the card of the previous player, he/she must put a card of the same suit of the previous card out. The player who put out the card with greater rank can take the two cards to his/her *scoring area*, and get the token, that is, to play first in the next round.
2. If not, he/she can put any card in hand. However, without considering the rank, the previous player can take the two cards to his/her *scoring area*. And the token stays still.

At the end, when players put all 13 cards out, a player's score is the number of HEART cards of his/her *scoring area*. The player with minor score wins. If the scores are the same, Alice wins.

Additional, we don't consider the "Shoot the Moon" of Spade♠ Q. Just follow the rules above.

Now, Alice and Bob are playing a *Microsoft Hearts* of 2 players. Both of them know the cards of each other. At the beginning, Alice has the token to put the card out first.

Alice wants to know, if both of them play the game optimally for themselves, whether she will win the game. Output "Yes" or "No".

Input

The first line contains a single positive integer n ($n \leq 13$), denoting the number of cards of one player.

The following two lines, each line contains n strings, denoting Alice's and Bob's cards, respectively. The string contains the rank and the suit. For example "7H" stands for ♥7, "TS" stands for ♠10.

Output

Print one single line.

If Alice wins, print "Yes". Otherwise, print "No".

Example

standard input	standard output
3 AH JH 7S 3H TD 5H	No

Problem H. coprime

Input file: `standard input`
Output file: `standard output`
Time limit: 1 second
Memory limit: 256 megabytes

How many positive integer sequences of length n satisfy that $1 \leq a_i \leq m$, and the adjacent elements are coprime.

The answer maybe too large, so you only need to output the result of its modulo of 998244353.

Input

Two positive ingeter $n(1 \leq n \leq 1000), m(1 \leq m \leq 1000)$.

Output

An integer represents the answer.

Example

standard input	standard output
2 5	19

Problem I. Bracket

Input file: **standard input**
Output file: **standard output**
Time limit: 2.5 seconds
Memory limit: 256 megabytes

Starlight Glimmer has a bracket sequence s of length n , which only contains character '(', ')' and '?'.

She also has a value sequence v of length n constructed by two integer constants a, b that $v_i = ai + b$.

Define that the partition of a sequence is a pair sequence (l_i, r_i) of length m if it satisfies $l_1 \leq r_1 < l_2 \leq r_2 < \dots < l_{m-1} \leq r_{m-1} < l_m \leq r_m$ and any substring (l_i, r_i) is a valid bracket sequence.

The substring (l, r) of s is a string removed first $l - 1$ characters and last $n - r$ characters of s , and Starlight Glimmer believes it valid if:

- Each character of it is either '(' or ')';
- The number of '(' is no less than the number of ')' for any prefix;
- The number of '(' is equal to the number of ')' for it.

According to the above, Starlight Glimmer defines the value of s is $\max_{\text{partition}} \sum v_{r_i+1-l_i}$.

Now Starlight Glimmer should and find the maximal value of s if she replace all '?' to '(' or ')' in a optimal way.

Input

The first line contains a single integer n ($1 \leq n \leq 5 \times 10^5$) - the length of s .

The second line contains a string s of length n - denoting the bracket sequence s .

The third line contains two integer a, b ($-10^6 \leq a, b \leq 10^6$) - denoting $v_i = ai + b$.

Output

Print a single integer - the maximal value of s after replacement.

Example

standard input	standard output
10 (?)(??(??)? 1 2	14

Note

For the testcase, $v = [3, 4, 5, 6, 7, 8, 9, 10, 11, 12]$.

The result of a possible replacement is '(()()())', and Starlight Glimmer could choose $(2, 3), (4, 5), (6, 9)$ as a partition, the value of it is $v_2 + v_2 + v_4 = 4 + 4 + 6 = 14$.

Problem J. intervals

Input file: **standard input**
Output file: **standard output**
Time limit: 2 seconds
Memory limit: 512 megabytes

You are given an array A with n integers. If an interval $[L, R]$ satisfies that the sequence $(A_L, A_{L+1}, \dots, A_R)$ has at least an odd number and at least an even number, we consider the interval $[L, R]$ as a perfect interval.

Now you are given q questions, each question contains an interval $[X, Y]$. You need to answer how many intervals $[L, R]$ satisfying $(X \leq L \leq R \leq Y)$ are perfect.

Input

The first line contains two integers n and q ($1 \leq n, q \leq 5 \cdot 10^5$), indicating the length of array A and the number of questions.

The second line contains n integers, indicating A_i ($0 \leq A_i \leq 10^9$).

From the third line to $(q + 2)$ -th line each line contains two integers X and Y ($1 \leq X \leq Y \leq n$), indicating the interval $[X, Y]$.

Output

The output has q lines, each line contains one integer, indicating the answer of each question.

Example

standard input	standard output
10 5	8
1 3 4 1 3 4 3 1 7 9	17
1 5	29
3 9	5
2 10	0
4 7	
1 2	

Problem K. Xor-permutation

Input file: `standard input`
Output file: `standard output`
Time limit: 1 second
Memory limit: 256 megabytes

Sunset Shimmer defines the value of a permutation p of length n is $\sum_{i=1}^n p_i \oplus i$, the $\oplus$ represents bitwise XOR.

Now she wants to know the maximal value of permutation of length n .

Input

The first line contains a single integer T ($1 \leq T \leq 10^5$) - the number of testcases.

Each testcase contains a single integer n ($1 \leq \sum n \leq 10^5$) - the length of the permutation.

Output

For each testcase, print one line of n integers, represents the maximal value permutation of length n .

If there are multiple solutions, just output any of them.

Example

standard input	standard output
2	3 1 2
3	5 1 4 3 2
5	

Problem L. FYC's luv letter

Input file: standard input
Output file: standard output
Time limit: 5 seconds
Memory limit: 1024 megabytes

FYC is a fucking cute boy who wants to write a beautiful love letter to the girl he likes. FYC knows a kind of character magic, and to show his sincerity, he first writes a love letter and then wishes to polish it with his own magic.

A love letter is a string, and a beautiful love letter must be a palindrome. FYC hopes that throughout the polishing process, his love letters will always be beautiful, that is, before using magic, during the process of being modified by magic, and after being modified by magic, love letters are always palindrome strings. The magic of FYC is carried out on the magic dictionary. There are n pages in the dictionary, and each page has a string. If the current love letter of FYC is the same as a certain substring of a certain page, then the love letter can be combined with the substring, and then modified by magic. FYC has the following three kinds of magic:

1. If the substring of the current string has two identical characters outside the two ends, you can spend a_{ij} costs to add these two identical characters to your love letter.
2. You can spend b_{ij} costs to delete the characters on both ends of the current love letter, but your love letter cannot be empty.
3. If in a certain position in the entire magic dictionary, there exists a substring that is the same as the substring you currently hold, you can spend $|i - j|$ costs to jump to the position of another substring, i and j are the two page numbers where the string is located. However, the distance of the page number between the target string and the current string cannot exceed k .

FYC has done some operations, but he doesn't know what to do next, and now he wants to ask for your help. That is, the love letter has now been placed on a certain substring on a certain page of the magic dictionary, and now it is hoped that you will use magic to transform the love letter into what he wants, and take it off a certain page. Please tell him what is the minimum cost he needs to spend.

Input

The first line n and k indicate a total of n strings and the range limit of operation 3;

The next $3n$ lines have the following format:

On the first line there is a string representing what the string on page i of the dictionary looks like. The length of the i^{th} string is m_i .

The second line contains 26 integer, indicating the cost of adding the two identical corresponding character ($a - z$) to the current page at the *operation1*.

The third line has 26 numbers, which represent the cost of deleting ($a - z$) the two identical corresponding character in the current page *operation2*.

A number on the next line represents the page number of the string where the starting string is located, and a string of characters represents the starting string;

The next line is a number indicating which page FYC wants to get the target string from, and a number indicating which page FYC wants to get the target string in the end;

It is guaranteed that all given letters are lower case letters.

$$1 \leq n \leq 100000$$

$$1 \leq m_i \leq 1000000$$

$$\sum m \leq 1000000$$

$$k \leq n$$

Problem M. Water

Input file: `standard input`
Output file: `standard output`
Time limit: 1 second
Memory limit: 256 megabytes

Now there are n people lining up to get some water, the i^{th} person wants to get c_i liters of water into his or her bottle.

When the bucket of water on the water dispenser runs out (Each bucket of water provides C liters of water) , the one who is using the water dispenser (includes the last one) will replace the bucket on the dispenser with a new bucket or water, even if he or she have got enough water. Then he or she leaves right away, so one may leaves with less water than one wants.

Now you want to know how many buckets of water needed (including the first bucket).

Input

The first line contains a single integer T ($1 \leq T \leq 100$) - the number of testcases.

Each testcase contains two lines.

The first line contains two integers n, C ($1 \leq \sum n \leq 10^6, 1 \leq C \leq 10^3$).

The second line contains n integers, c_i ($1 \leq c_i \leq 10^3$).

Output

Print a positive integer representing the number of buckets needed.

Example

standard input	standard output
4	4
3 1	5
6 4 5	1
4 1	3
1 6 4 5	
2 10	
5 2	
5 10	
8 6 7 10 2	